

RRI Recap + Policy Brief

The **AREA** framework for RRI

Anticipate, Reflect, Engage, Act*

▶ Developed in the UK by the Engineering and Physical Sciences Research Council based on foundational work by sociologists of science and technology Jack Stilgoe, Richard Own and Phil Macnaghten in the late 00s.

▶ Four key dimensions:

- **Anticipate:** Explore possible impacts on society, environment, etc., intended and unintended, avoid restricting to what can be predicted.
- **Reflect:** Consider purpose of and motivation for research, assumptions, associated uncertainties, areas of ignorance, framings, etc.
- **Engage:** Open the above discussions on visions and impacts to broader deliberation through dialogue with relevant societal stakeholders.
- **Act:** Use these processes to influence the direction and trajectory of the research and innovation process itself. Why is that hard? How to do it?

**Originally anticipation, reflexivity, inclusion, and responsiveness in the foundational research policy paper by Stilgoe et al (2013)*

Anticipate

- ▶ Take uncertainty seriously: It is hard to predict the effects of emerging technologies.
- ▶ The goal is to explore/imagine as many **possible** intended and unintended consequences of the research/innovation as possible. **As opposed to** establishing with certainty the probability of only the consequences that we understand well.
- ▶ Typical impulse of science is to narrow the imagination around impacts to what falls within area of expertise. Anticipation must work against that impulse.
- ▶ Comfortable questions within area of expertise: do we have the appropriate methodology, risk assessment, data management plan, lab health and safety, ethical approval, informed consent?
- ▶ Expand to questions like:
 - is this socially desirable?
 - for which stakeholders?
 - in what scenarios?
 - addressing what kind of larger societal challenges?

Reflect

▶ Science is already good at reflecting - it is organized scepticism (Popper).

▶ In this context: reflect on the consequences of own assumptions, of the way we have chosen to frame the problem, of our motivations for doing the research.

▶ Expand to:

- Which groups or individuals have a stake in the research and its outcomes?
- Which groups or individuals influence the research? In what ways do these stakeholders bring their influence to bear?
- Which groups or individuals can be held responsible in any way for unwanted/harmful consequences of the research?

▶ How do you know the answer to these questions? Who decides who the right stakeholders are? How do we decide what kind of stakes they have and who should be held accountable?

Engage

- ▶ Invite societal stakeholders to participate in your anticipation and reflection.
- ▶ Do they begin from the same assumptions as you? Do they see the same purpose and motivation for doing the research as you? Do they frame the problems to be solved in the same way as you?
- ▶ Do they agree with you on who the relevant people or groups are to invite, i.e. do they agree on who has a stake?
- ▶ Do they agree on who should be held accountable? Do they agree with you on who has influence on the research?
- ▶ Can they imagine other intended or unintended consequences of the research/innovation than you? Can they imagine other scenarios?
- ▶ Do they agree with you on what is socially acceptable/desirable?

Act

How do we do this?

What will it require in practice?

- ▶ Interdisciplinary training and collaboration. Basically why you are doing this course.
- ▶ Methods for technology assessment, scenario scanning, public engagement and deliberation, co-design, etc. On this course you will learn to:
 - Conduct sustainability assessment (Part 1)
 - Map and analyse public controversies around technology and innovation (Part 2)
 - Organize public engagement exercises (Part 2)
 - Apply design methods with diverse stakeholders in mind (Part 2)
- ▶ Creating structural incentives. Refer back to the way the current funding system for research and innovation incorporates RRI principles.

RRI and your policy brief in short

- ▶ You are writing your policy brief for a client (decision maker) that will implement your proposed sustainability improvements (you decide who). This gives the focus for your policy brief. Your proposal can come from Assignment B (Part 1) of the course.
- ▶ The objective is to help your client understand the energy island controversy, who the actors are and what they are concerned about, and develop an RRI strategy for your proposal.
- ▶ RRI is about anticipating how your innovation can be a problem in society. To do so, you reflect on (the limits of) your own assumptions by engaging stakeholders who see it differently.
- ▶ You must first quantitatively map the actors and issues using the methods and concepts of controversy mapping you learned in the mapping lectures of Part 2. Your **methodology** section shows your client how you have chosen to focus on certain actor statements and why.
- ▶ You must then qualitatively analyse those actor statements. Here you will need the analytical concepts you learned in the other lectures on Part 2 of the course. Your **analysis** helps your client understand what the actors are actually discussing and whether they have different risk perceptions, ethical frameworks, ideas about expertise, users, markets, etc.
- ▶ Based on your findings, suggest to your client who could be engaged to help you reflect on your assumptions in what way and anticipate what kind of adverse consequences how. This is your **recommendation** for an RRI strategy. It should be justified by one of the four dimensions.

The Concept Glossary

- ▶ The Concept Glossary is a **learning resource** that contains all concepts you learned in Part 2 of the course. It is a definitive list of concepts you can use in the policy brief. For the MCT, you should also the course materials of Part 1, as well as the lecture notes and slides of Part 2.
- ▶ A concept is a **thinking tool**: an idea that helps you notice, describe, and reason about aspects of the world. You can think of a concept as a lens: it makes you see some features, while de-emphasising other features. This course distinguishes three types of concepts:
- ▶ **Methodological concepts** guide how you work with data. In this course, they help you parse and analyse a dataset of statements following a controversy mapping approach. Methodological concepts provide rules of thumb and quality criteria: what to look for, what to ignore, how to justify your choices, and how to make your analysis robust.
- ▶ **Analytical concepts** guide what you pay attention to in the phenomenon you study. They help you formulate the questions you ask and make explicit which dimensions you foreground. In other words: they shape your particular analytical angle. You should have **ONE analytical focus** per analysis section, for example, ethics, markets, or risk perception.
- ▶ **Interventional concepts** guide what you propose to do based on the analysis. They help you develop, justify, and communicate recommendations in your policy brief, i.e., how to respond to the issues you identified in the analysis. You should use **ONE concept** per recommendation.

Monday 12/1	Tuesday 13/1	Wednesday 14/1	Thursday 15/1	Friday 16/1
Responsible Research and Innovation (RRI)	Sociology of science and expertise	Controversy mapping introduction	Social acceptance	Markets in the making
Ethics	Risk and uncertainty	Mapping #1: Explore the issue atlas	Mapping #3: Making data-driven points	Universal design
The politics of technology	Mandatory assignment #C RRI in your field	Mapping #2: Follow the actors	Mapping #4: Visual storytelling	Mandatory assignment #D Issue atlas driver's license

Monday 19/1	Tuesday 20/1	Wednesday 21/1	Thursday 22/1	Friday 23/1
Users, innovation and resistance	Write policy brief	Mandatory assignment #F Policy conference	Write policy brief	MULTIPLE-CHOICE TEST
Public engagement	Write policy brief	Write policy brief	Write policy brief	Write policy brief
RRI recap and policy brief intro	Q&A based supervision	Q&A based supervision (online)	Write policy brief	HAND IN POLICY BRIEF

Assignment E: Policy conference

▶ Task: In this assignment, your group must **present a plan for your policy brief** to two other groups and **receive feedback**. You will also provide feedback on the presentations of those two groups. To pass the assignment, you must submit your group's presentation as PDF and fill in the feedback boxes.

▶ Format of presentation: Introduction (1 slide), Methodology (1 slide), Analysis (1 slides), Recommendation (1 slide)

▶ On **Wednesday 21 January**, all students will join Zoom **at 10:15** using the **general zoom link** for the course provided on Learn. You should join the breakout room that is assigned to your group (see schedule in the assignment document).

▶ You agree on an order of presentation with your fellow students. Each group present their plan for the policy brief in 4 slides (max. 5 min.) The two respondent groups discuss the presentation and how you would assess it based on the evaluation criteria for the policy brief assignment (max. 5 mins.).

▶ Give **constructive feedback**! For example, begin by stating what you liked about the presentation and the recommendations. Maybe it made you think about something in a new way or surprised you. Then assess how it lives up to the **evaluation criteria** for the policy brief assignment (see description) and where you see room for improvement.

Live Q&A